Industrial Internship Report

TECH ELECON

Submitted by

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12102120601016

In partial fulfillment for the award of the degree of

Bachelor of Technology

in

Artificial Intelligence & Data Science

A. D. Patel Institute of Technology

The Charutar Vidya Mandal (CVM) University

Vallabh Vidyanagar - 388120

February 2025

A. D. Patel Institute of Technology

Artificial Intelligence & Data Science

CERTIFICATE

This is to certify that **PATEL DHRUV VASANTKUMAR (12102120601016)** has submitted the Industrial Internship report based on internship undergone at **Tech Elecon** for a period of **16** weeks from **Jan 1 2025** to **April 30 2025** in partial fulfillment for the degree of Bachelor of Technology in **Artificial Intelligence & Data Science**, **A. D. Patel Institute of Technology** at The Charutar Vidya Mandal (CVM) University, Vallabh Vidyanagar during the academic year 2025 – 26.

Signature:
Name: Prof. Mayur Ajmeri
Internal Guide

Signature:
Name: Dr. Dinesh Prajapati
Head of the Department

TECHELECON
TECHNOLOGY SIMPLIFIED

HR/TEPL/110 **10.12.2024**

To,
A D Patel Institute of Technology,
V V Nagar.

Permission for Internship

Dear Sir/M'am,

With reference to your letter dated.30.11.2024, we are pleased to grant permission to ***Mr.Dhruv Vasantkumar Patel*** *student of* ***Artificial Intelligence*** *for* ***Internship*** in our organization from ***01.01.2025 to 30.04.2025.***

The following points are to be adhered by the student.

- Laptops, Pen-Drives & other hard drives are not permitted inside the company premises.
- Appropriate dressing & grooming will be appreciated. Jeans & T Shirts are not allowed.
- Female students have to wear dress and Kurtis only.
- Photography & Videography is strictly prohibited in the premises.
- Trainee/s has to strictly comply with the Company's timings & schedule.
- In case of any casualty the Company will not be held responsible.
- On Completion of training the trainee/s has to submit a copy of Training Report duly signed by concern guide to obtain the Training Certificate.
- In case of any ambiguity / difficulty, the trainee is required to approach HR Department. The detailed guideline are attached overleaf.

For, Tech Elecon Pvt.Ltd.

Nirali Trivedi
Group HR Head
Cc. Satyam Raval

DECLARATION

I, Patel Dhruv Vasantkumar (12102120601016), hereby declare that the Industrial Internship report submitted in partial fulfillment for the degree of Bachelor of Technology in Artificial Intelligence & Data Science , A. D. Patel Institute of Technology, The Charutar Vidya Mandal (CVM) University, Vallabh Vidyanagar, is a bonafide record of work carried out by me at Tech Elecon under the supervision of Satyam Raval and Prof. Mayur Ajmeri and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of the Student — Signature of Student

Patel Dhruv Vasantkumar — ________________

Acknowledgement

I would like to express my sincere gratitude to **Tech Elecon Pvt. Ltd., Anand, Gujarat,** for providing me with the opportunity to complete a **four-month internship in Data Science.** This experience has been invaluable in enhancing my technical skills and knowledge in the field of Data Science, AI and chatbot development.

I am deeply thankful to **Prof. Mayur Ajmeri, my Internal Guide,** for his continuous support, guidance, and mentorship throughout the internship. His insights and expertise were instrumental in helping me understand and apply key concepts in AI/ML and Data Science.

I also extend my appreciation to the **entire team at Tech Elecon** for creating a collaborative and encouraging learning environment. Their professionalism and willingness to share knowledge have significantly contributed to my learning experience.

This internship has been a significant step in my professional journey, and I am grateful for the opportunity.

Thank you.

Abstract

This report presents a comprehensive overview of the experiences and learnings acquired during a **four-month internship** at **Tech Elecon Pvt. Ltd**., Anand, Gujarat, in the domain of **Data Science**. The internship was designed to provide practical exposure to AI/ML concepts, model development, natural language processing (NLP), and chatbot implementation, with a focus on enhancing organizational efficiency. Throughout this period, I engaged in various tasks related to machine learning, including data preprocessing, model testing, and deploying AI solutions tailored for HR applications.

The flagship project, **"HRMS Chatbot",** is an AI-powered chatbot developed to assist employees with HR-related queries, thereby streamlining internal communication within the organization. This locally hosted solution enables users to retrieve policy documents and address common HR inquiries efficiently while ensuring strict data confidentiality, a critical requirement for sensitive HR information. The project employs Retrieval-Augmented Generation (RAG) for response accuracy, utilizing ChromaDB as a local vector store for the company handbook and Ollama for hosting lightweight NLP models. A user-friendly web interface was crafted using Streamlit, incorporating features such as chat history, timestamps, and export options. This report encapsulates the learning journey, detailed project development, challenges encountered, and the valuable insights gained over the four-month internship.

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Abbreviations

- AI – Artificial Intelligence
- ML – Machine Learning
- NLP – Natural Language Processing
- DL – Deep Learning
- API – Application Programming Interface
- GPU – Graphics Processing Unit
- AWS – Amazon Web Services
- UI – User Interface

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1. OVERVIEW OF THE COMPANY

1.1. HISTORY

- **Tech Elecon Private Limited**, established on May 30, 2012, is an IT services company headquartered in Vallabh Vidyanagar, Gujarat, India. It specializes in **internet communication, hardware maintenance, and professional security solutions**. The company operates as a subsidiary of Elecon Engineering Company Limited, a leading multinational in industrial gears and material handling equipment. With a registered office at Anand Sojitra Road, its board of directors includes **Pradip Manubhai Patel, Prayasvin Patel, and Ravin Shah**. Financially, Tech Elecon has shown strong growth, reporting a 23.5% increase in revenue and a remarkable 461.38% rise in net profit for the fiscal year ending March 31, 2023. The company's net worth also saw a substantial 181.83% growth during this period. With a solid foundation and access to industry expertise, Tech Elecon continues to expand its presence in the IT sector, offering innovative solutions to clients. Over the years, it has built a reputation for reliability and technological advancement in its field.

1.2. SCOPE OF WORK

Tech Elecon offers a **comprehensive range of technology services** designed to empower businesses through digital transformation. Their core service lines include:

- **Industrial IoT Monitoring System:** Tech Elecon has developed an advanced Industrial Internet of Things (IIoT) monitoring system designed for manufacturing and engineering industries. This system collects real-time data from connected machines and sensors, providing businesses with valuable insights into equipment performance, predictive maintenance, and production efficiency. By leveraging IIoT, industries can minimize downtime, optimize resource utilization, and enhance overall productivity.

- **Cybersecurity & Network Protection Suite:** To ensure data security and prevent cyber threats, Tech Elecon offers a comprehensive cybersecurity and network protection suite. This solution includes firewall management, intrusion detection, data encryption, and vulnerability assessments. Designed for enterprises and government organizations, this suite helps businesses safeguard sensitive information, comply with industry regulations, and maintain a secure IT infrastructure.

- **Cloud-Based ERP System:** Tech Elecon provides a cloud-based Enterprise Resource Planning (ERP) system that streamlines business operations by integrating various departments such as finance, supply chain, human resources, and customer management. The system allows companies to access real-time data, automate workflows, and improve decision-making processes. With its scalable architecture, this ERP solution is suitable for businesses of all sizes looking to enhance operational efficiency.

- **AI-powered IT Helpdesk Automation:** To improve IT support efficiency, Tech Elecon has developed an AI-powered helpdesk automation system. This system leverages machine learning and natural language processing (NLP) to handle IT service requests, troubleshoot technical issues, and assist employees with common IT concerns. By reducing response times and automating routine tasks, the solution enhances productivity and minimizes IT downtime.

- **Smart Energy Management Solution:** Tech Elecon has introduced a smart energy management system that helps organizations monitor and optimize their energy consumption. By integrating IoT sensors and AI-driven analytics, the system provides real-time insights into power usage patterns, identifies inefficiencies, and suggests energy-saving measures. This solution is particularly beneficial for industries and commercial buildings aiming to reduce operational costs and improve sustainability.

1.3. ORGANIZATION CHART

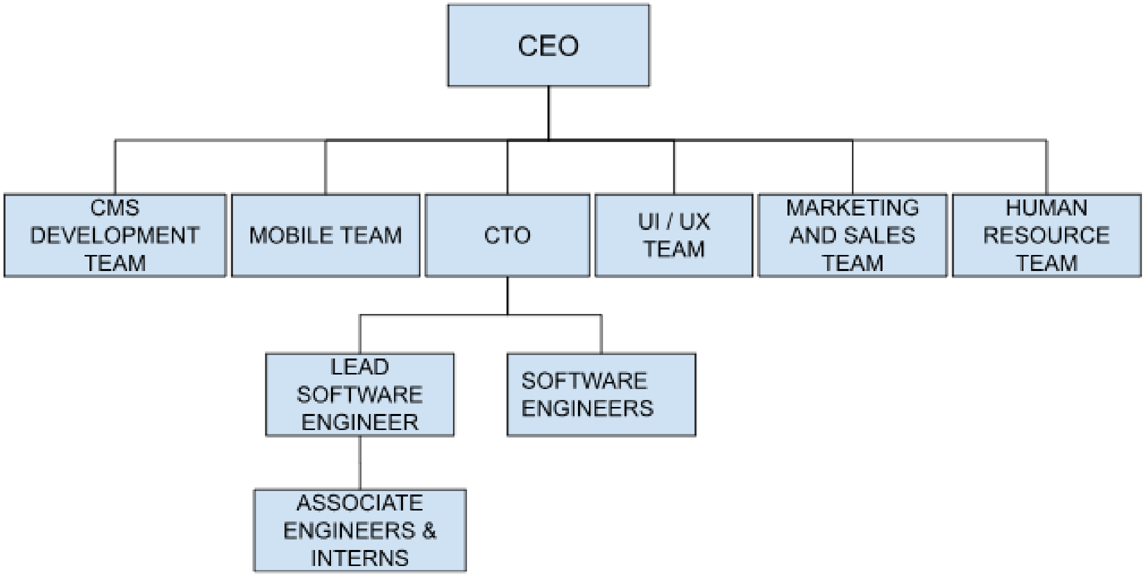

Figure 1.1 Organization char

2. INTERNSHIP AND PROJECT

2.1. INTERNSHIP SUMMARY

2.2. PURPOSE

The purpose of this project was to develop an AI-powered HR assistant that operates entirely locally to streamline employee interactions with HR services while safeguarding the confidentiality of company policies. As organizations increasingly seek automated solutions to improve HR efficiency, this chatbot was designed with specific goals in mind:

- Reduce dependency on HR personnel for repetitive queries by providing instant access to policy-related information.
- Enhance accessibility for employees seeking quick, secure answers to HR questions directly from the company handbook.
- Improve operational efficiency in handling routine HR tasks, laying the groundwork for potential future enhancements such as leave management or document processing.

2.3. OBJECTIVE

The primary objectives of the HRMS Chatbot project were multifaceted, aiming to deliver a robust and secure solution:

- Develop an AI chatbot capable of efficiently handling HR-related queries using locally hosted resources to ensure data privacy.
- Implement NLP models to accurately interpret employee requests and retrieve relevant answers from the company handbook.
- Ensure a secure and compliant system by maintaining all operations locally, protecting sensitive employee and organizational data.
- Deploy the chatbot as a web-based application with an intuitive interface to facilitate seamless user interaction.

2.4. SCOPE

The HRMS Chatbot project primarily focuses on delivering a targeted set of functionalities within a secure, local environment:

- Providing instant responses to employee queries regarding HR policies, procedures, and guidelines sourced from the company handbook.
- Facilitating secure retrieval of policy information through a locally deployed vector database, ensuring no external data exposure.
- Deployment as a web-based chatbot with a Streamlit interface, designed for potential future integration with broader HR systems.

The scope does not encompass handling confidential HR decision-making processes or replacing human HR personnel for complex cases. Instead, the system functions strictly as an AI-powered assistant to augment HR service efficiency, prioritizing data confidentiality and operational simplicity.

2.5. TECHNOLOGY AND LITERATURE REVIEW

<u>Technology will be using:</u>

Table 2.1 Tech Stack will be using

Technology	Purpose
Python	Backend scripting and integration for chatbot logic
Ollama	Local hosting and inference of lightweight NLP models
ChromaDB	Local vector store for storing and retrieving handbook embeddings
Streamlit	Development of an interactive web-based chatbot interface
nomic-embed-text	Generating text embeddings for handbook data processing

Literature review:

Recent advancements in AI and NLP have significantly transformed human resource management, improving efficiency in employee interactions, query resolution, and administrative task automation. AI-powered chatbots in HRMS have been shown to substantially reduce manual workloads by providing automated responses to **frequently asked questions and policy-related inquiries [1]**. Research underscores their capability to enhance employee engagement and offer 24/7 HR assistance, making them indispensable in **modern organizational settings [2]**.

The adoption of local deployments for AI systems has gained traction, particularly for handling sensitive data, as it aligns with privacy regulations such as GDPR and ISO standards, ensuring **secure and transparent management of employee information [3]**. Lightweight NLP models, typically ranging from 1 to 1.5 billion parameters, have proven effective for local inference on standard hardware, offering a practical solution for confidential **HR applications without requiring extensive computational resources [4]**. However, implementing such systems presents challenges, including optimizing response accuracy and managing performance on **CPU-based environments, necessitating careful model selection and prompt engineering [5]**.

Continuous improvement through techniques like prompt tuning and comparative model testing is recommended to enhance chatbot performance in domain-specific contexts, such as **HR policy retrieval [6]**. In summary, locally deployed AI chatbots provide a balanced approach to efficiency, security, and scalability for HR automation, provided organizations address **technical constraints and maintain a focus on data privacy [7]**.

2.6. PROJECT PLANNING

2.6.1. PROJECT DEVELOPMENT APPROACH AND JUSTIFICATION

The **Agile methodology** will be used for developing the HRMS chatbot, ensuring an **iterative development process** with continuous feedback, sprint-based enhancements, and flexibility to incorporate evolving HR requirements. Agile has been widely adopted in AI-driven enterprise solutions due to its **scalability, efficiency, and adaptability**.

The chatbot will integrate **Natural Language Processing (NLP) and Retrieval-Augmented Generation (RAG)** to handle employee queries, document retrieval, and HR-related tasks efficiently. Studies suggest that NLP-powered HR chatbots **improve query resolution speed, enhance employee engagement, and automate routine HR tasks** with high accuracy.

The choice of lightweight models, such as those with **1-1.5 billion parameters**, was justified by their ability to perform effectively on CPU hardware, aligning with the project's resource constraints and the need for a **fully local deployment to protect sensitive HR data**. This approach balances performance with practicality, making it suitable for real-world HR applications where data security is paramount.

2.6.2. COST ESTIMATION

To optimize costs while maintaining system efficiency, **open-source technologies, free-tier cloud services, and efficient resource allocation** will be utilized. Below is the cost estimation for the HRMS chatbot development:

Table 2.2 Cost Estimation

Components	Description	Estimated Cost (Rs)
Computing Resources	Local CPU-based laptop (existing)	0
Vector Database	ChromaDB (open-source vector store)	0
AI Model Usage	Ollama with locally hosted models	0
Development Tools	Open-source libraries (Streamlit, Python, etc.)	0

2.7. PROJECT SCHEDULING

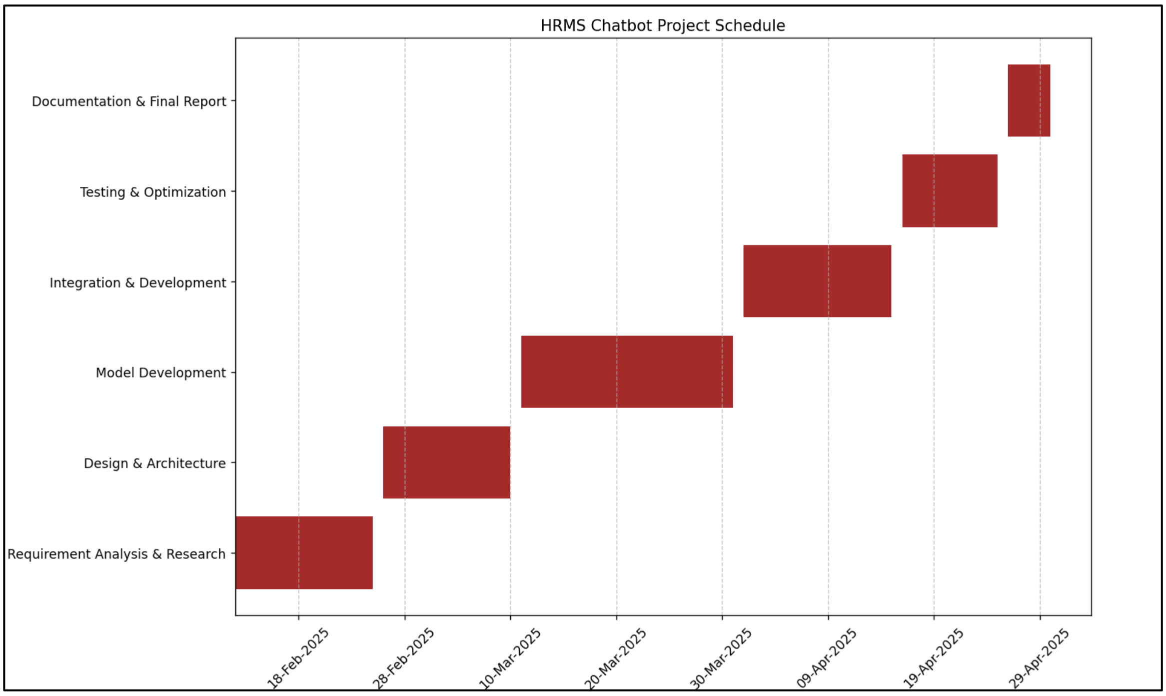

Figure 2.1 Gannt chart

3. SYSTEM ANALYSIS

3.1. STUDY OF CURRENT SYSTEM.

Currently, HR-related queries and employee management tasks at Tech Elecon Pvt. Ltd. are primarily managed through manual processes or standalone HR software systems. While these methods are operational, they present several limitations, including delays in response times, lack of tailored assistance, and restricted accessibility outside regular office hours. Employees typically depend on HR personnel or internal documentation, which often fails to deliver prompt and precise answers to their inquiries.

Existing HR chatbots have limitations, such as:

- **Lack of contextual understanding** – Many chatbots struggle to interpret intricate employee queries related to policies, benefits, or procedural guidelines [8].
- **Rule-based constraints** – Some rely on rigid, predefined question-answer frameworks, limiting their adaptability to diverse or nuanced requests [9].
- **Data privacy vulnerabilities** – Security of employee data and compliance with regulations (e.g., GDPR, labor laws) are not consistently guaranteed.
- **Limited integration capabilities** – Many fail to seamlessly connect with enterprise HRMS platforms, reducing their overall utility.

Despite progress in AI and Natural Language Processing (NLP), current HR chatbots often fall short in delivering accurate, personalized, and secure responses tailored to organizational needs.

3.2. PROBLEM AND WEAKNESS OF CURRENT SYSTEMS

The major shortcomings of existing HR management methods and chatbots include:

1. **Delayed Responses & Limited Availability**

   - Traditional HR support requires employees to wait for HR personnel availability, causing delays in query resolution [10].
   - Queries raised outside working hours often remain unresolved until the next business day.

2. **Lack of Personalization**

   - Existing HR bots provide generic responses instead of personalized answers based on employee records.
   - They do not track employee history, benefits, or department-specific policies to provide relevant guidance.

3. **Difficulty in Policy Understanding**

   - Employees often struggle to navigate complex HR policies, leave regulations, and payroll structures.
   - HR portals contain vast documentation, making it hard to find precise information quickly.

4. **Data Security & Compliance Issues**

   - Many HR chatbots do not fully adhere to data protection standards, posing risks to sensitive employee information [11].
   - Sensitive employee information, such as payroll and personal details, must be protected under GDPR and company regulations.

5. **Limited Integration with HRMS**

   - Existing chatbots often do not sync well with HR systems like leave management, payroll, and employee feedback portals.

- This leads to repetitive data entry and inefficiencies in employee self-service functions.

3.3. REQUIREMENTS OF NEW SYSTEMS

To overcome these limitations, the new AI-powered **HRMS Chatbot** will incorporate:

1. **Smart AI-Powered Responses**
   - Utilize **advanced NLP models** to understand and respond to HR-related queries more accurately.
   - Train the AI on **company-specific HR policies** for precise and context-aware answers.
2. **Personalized Assistance**
   - Track **employee history, roles, and benefits** to offer customized responses.
   - Adapt chatbot responses based on past interactions and frequently asked questions.
3. **24/7 Availability & Instant Support**
   - Provide **round-the-clock HR assistance** to employees across different time zones.
   - Reduce dependency on HR personnel for routine queries like leave balance, payroll information, and policy clarifications.
4. **Seamless Integration with HRMS & Other Tools**
   - Connect with existing **HR platforms (SAP, Workday, Oracle HRMS, etc.)** for real-time data access.
   - Automate tasks such as **leave applications, payroll inquiries, and employee feedback collection**.
5. **Secure Data Handling & Compliance**
   - Ensure compliance with **GDPR, labor laws, and company data protection policies**.

- Implement **end-to-end encryption** to protect sensitive employee records and interactions.

6. **Multi-Platform & Multi-Language Support**

- Deploy the chatbot as a web-based solution with potential expansion to platforms like **Slack** or **Microsoft Teams**.
- Plan for **multilingual capabilities** to accommodate a diverse workforce in future iterations.

By addressing these critical gaps, the new **HRMS chatbot** aims to improve **employee experience, efficiency, and HR automation**, making HR processes smoother and more effective.

3.4. SYSTEM FEASIBILITY

3.4.1. DOES THE SYSTEM CONTRIBUTE TO THE OVERALL OBJECTIVES OF THE ORGANIZATION

System feasibility assesses whether the proposed AI-based HRMS chatbot can be successfully developed and implemented within Tech Elecon under technical, financial, and operational constraints. Given that the chatbot is an AI/ML-focused project, the feasibility study ensures that the system:

- Supports organizational objectives by streamlining HR processes, improving employee access to information, and boosting efficiency.
- Utilizes cost-effective, open-source technologies to minimize expenses.
- Can be developed within the internship timeline using existing resources.
- Offers potential for integration with current HR systems, enhancing its value to the organization.

3.4.2. CAN THE SYSTEM BE IMPLEMENTED USING THE CURRENT TECHNOLOGY AND WITHIN THE GIVEN COST AND SCHEDULE CONSTRAINTS

- **Technical Feasibility**

The chatbot will be developed using **free and open-source tools**, ensuring minimal costs

Table 3.1 Technical Feasibility

Component	Technology used	Remarks
Frontend	Streamlit / HTML	Lightweight, user-friendly for rapid UI development
Backend/Logic	Python	Versatile language for AI/ML and application logic
LLM	Ollama	Local hosting of lightweight models (e.g., 1-1.5B parameters)
Embedding Model	nomic-embed-text	Efficient text embedding for handbook data
Vector Database	ChromaDB	Open-source, local storage for similarity search
Document Storage	Local File System	Simple and secure for initial development
Deployment	Local Deployment	Fully local setup for confidentiality
Data	HR Documents (PDFs)	Specific HR data used for training and retrieval.

- **Economic Feasibility**

Since the goal is to keep **costs minimal**, the following cost-effective strategy is adopted:

Table 3.2 Economic Feasibility

Expense	Estimated Cost (Rs)	Remarks
Computing Resources	0	Uses existing CPU-based laptop
Database Storage	0	ChromaDB (open-source)
AI Model Usage	0	Ollama with local models
Development Tools	0	Open-source libraries (Streamlit, etc.)
Total	**0**	**No additional costs incurred**

- **Schedule Feasibility**

Table 3.3 Schedule Feasibility

Phase	Timeline
Requirement Analysis & Planning	1 week
UI Design & Prototyping	1 week
Backend & AI Model Integration	3 week
Training & Optimization	2 week
Testing & Debugging	2 week
Deployment	1 week

3.4.3. CAN THE SYSTEM BE INTEGRATED WITH OTHER SYSTEM WHICH ARE ALREADY IN PLACE?

Integration with existing HR systems will enhance the chatbot's value. The chatbot will be designed to support:

- Document Management Systems: Integration with existing systems for seamless document handling.
- Employee Databases: Access to employee data (with appropriate permissions) for personalized responses and HR functions.
- Communication Platforms: Potential integration with Slack, Microsoft Teams, or other platforms for broader accessibility.

3.5. PROCESS IN NEW SYSTEM

The HRMS chatbot will follow a structured process:

- **User Inquiry**: Employee submits an HR-related question via the Streamlit interface.
- **NLP-Based Understanding**: Ollama’s NLP models analyze the query locally.
- **Information Retrieval**: ChromaDB retrieves relevant handbook data using nomic-embed-text embeddings.
- **Response Generation**: The selected model (e.g., llama3.2:1b) generates a response based on the query and context.
- **Display**: The response is presented in the Streamlit UI with chat history and interactive options.
- **Logging**: Interactions are stored locally for analysis and improvement.

3.6. FEATURES OF NEW SYSTEMS

- **AI-Powered HR Chatbot:** Utilizes local NLP models for intelligent, handbook-based responses.
- **24/7 Availability:** Offers continuous support via a web interface.
- **Localized Responses:** Tailored to company handbook content.
- **Self-Service Interface:** Includes chat history, timestamps, and export options.
- **Secure Data Handling:** Fully local deployment ensures confidentiality.
- **Scalability:** Designed for potential expansion with additional features.

3.7. SELECTION OF HARDWARE, SOFTWARE AND ALGORITHMS

Hardware Selection:

- **Development Environment:** Standard laptop (16GB RAM, quad-core CPU).
- **Deployment Environment:** Local server (initially), with potential for enhanced hardware in future phases.

Software Selection:

- **Operating System:** Windows/Linux (development).
- **Programming Language:** Python.
- **Frameworks:** Streamlit, Ollama.
- **Vector Database:** ChromaDB.
- **Embedding Model:** nomic-embed-text:latest.
- **Version Control:** Git, GitHub.

Algorithm Selection:

- **LLM:** Lightweight models via Ollama (e.g., qwen2.5:1.5b, deepseek-r1:1.5b, gemma3:1b, llama3.2:1b).
- **Embedding Model:** nomic-embed-text:latest for text representation.
- **Vector Search:** ChromaDB’s similarity search algorithms

4. SYSTEM DESIGN

This section outlines the overall system architecture, design methodology, process flows, and the associated UML diagrams that help visualize the system's structure and behavior.

4.1. SYSTEM DESIGN AND METHODOLOGY

The design of the health chatbot is based on a **layered, modular architecture** that facilitates ease of development, scalability, and integration with existing systems. The main layers include:

- **Presentation Layer (Frontend):**
  - **Technologies:** Streamlit or a lightweight web interface using HTML and CSS.
  - **Function:** Provides an intuitive web-based interface for employees to interact with the chatbot, featuring chat input, history display, timestamps, and export options.
- **Application Layer (Backend/API):**
  - **Technologies:** Python.
  - **Function:** Manages the core logic, processes user queries, and coordinates between the frontend and AI components, ensuring seamless operation within a local environment.
- **AI/ML Processing Layer:**
  - **Technologies:** Ollama with lightweight models (qwen2.5:1.5b, deepseek-r1:1.5b, gemma3:1b, llama3.2:1b).
  - **Function:** Implements Natural Language Processing (NLP) for query understanding and response generation, leveraging Retrieval-Augmented Generation (RAG) with company handbook data.

- **Data Storage Layer:**
  - **Technologies:** ChromaDB.
  - **Function:** Stores handbook embeddings locally using nomic-embed-text:latest, enabling fast similarity searches and secure data management without external dependencies.
- **Integration Layer (Future Potential):**
  - **Technologies:** Planned for RESTful APIs.
  - **Function:** Facilitates potential future integration with internal HR systems (e.g., SAP, Workday) for enhanced functionality, while maintaining a local-first approach.

Methodology:

The development adheres to an **Agile methodology**, characterized by iterative sprints and continuous refinement. This approach ensures:

- **User requirements** are iteratively clarified and addressed, focusing on HR-specific needs like policy retrieval.
- **Incremental improvements** are implemented based on testing results and model performance evaluations.
- **Flexibility** is preserved to adapt to additional features or optimizations as the project progresses within the internship timeline.

4.2. PROCESS DESIGN

The process design details the step-by-step flow of how the chatbot handles user interactions and generates responses. The key stages include:

- **User Input and Query Submission:**

  Action: The employee enters an HR-related query (e.g., "What's the leave policy?") via the Streamlit interface.

  Output: The query is transmitted to the backend Python logic for processing.

- **Pre-processing and NLP Analysis:**

  Action: The backend processes the input using Ollama's NLP capabilities.

  Processing:

  - **Query Understanding:** The selected model (e.g., llama3.2:1b) interprets the intent and structure of the query.
  - **Context Preparation:** The query is prepared for retrieval without additional tokenization tools, relying on the model's built-in capabilities.

- **Information Retrieval and Response Generation:**

  Action: ChromaDB retrieves relevant handbook sections using nomic-embed-text:latest embeddings, and Ollama generates a response.

  Processing: The model combines the retrieved context with the query to produce a concise, handbook-based answer, ensuring accuracy and relevance.

- **Response Delivery and Logging:**

  Action: The response is displayed in the Streamlit interface with chat history and interactive features.

  Logging: Interactions are logged locally in the system for performance analysis and future refinements, maintaining data privacy.

- **Evaluation and Iterative Improvement:**

  Action: Model performance is assessed through testing with sample queries (e.g., "What's the dress code?").

  Processing: Insights from response times and quality guide prompt tuning and model comparisons for ongoing enhancement.

4.3. UML DIAGRAMS

- **Use Case Diagram:**

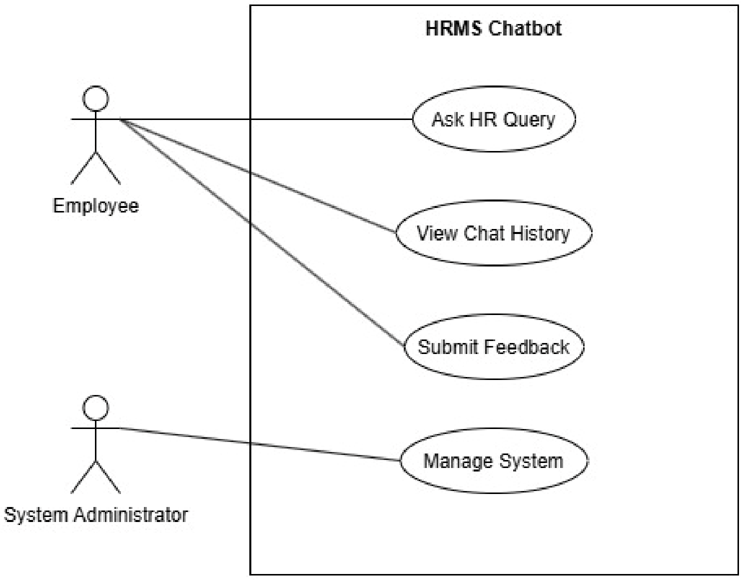

Figure 4.1 Use case Diagram

- **Sequence Diagram:**

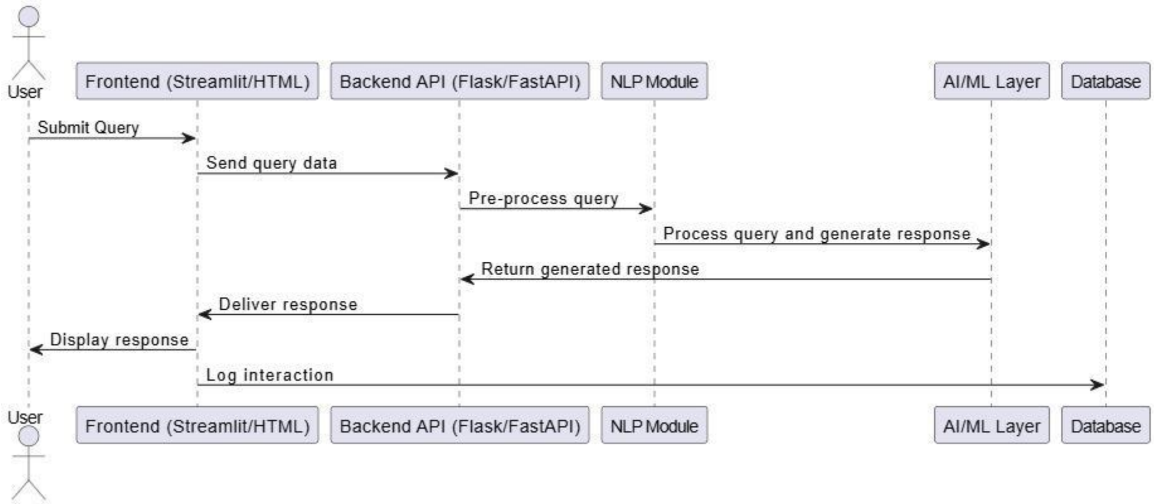

Figure 4.2 Sequence Diagram

- **Class Diagram:**

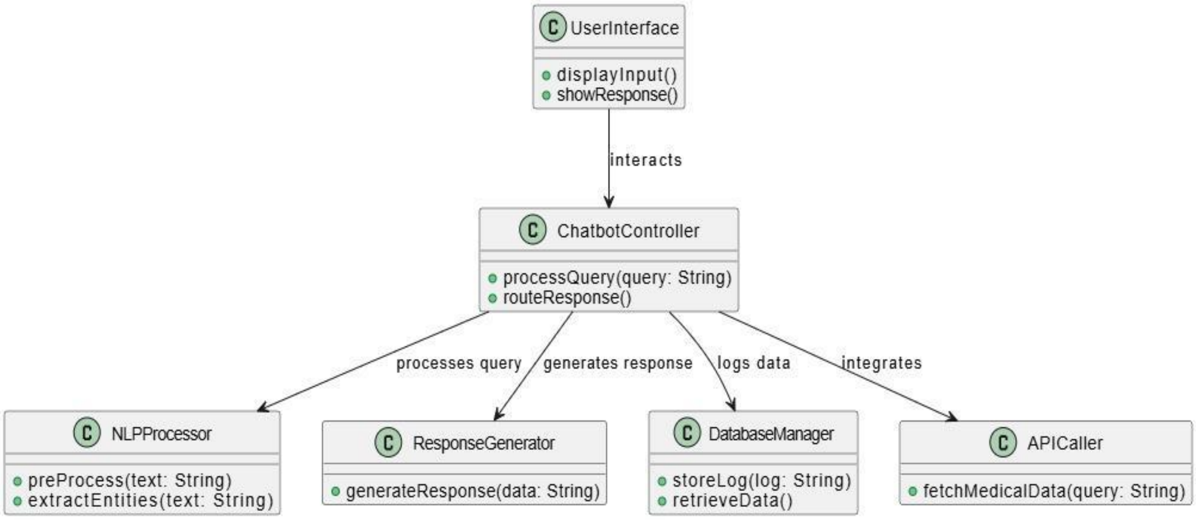

Figure 4.3 Class Diagram

- **Activity Diagram:**

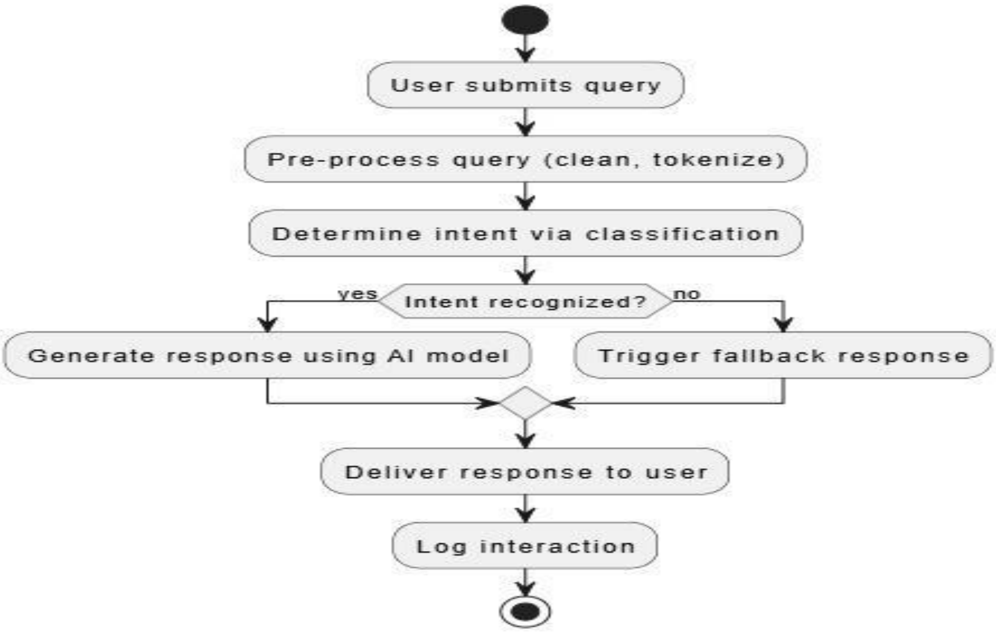

Figure 4.4 Activity Diagram

5. IMPLEMENTATION

This section details the practical execution of the HRMS chatbot project, encompassing the platform and environment used, the processes and technologies employed, the findings and outcomes achieved, and an analysis of the results with comparisons and future deliberations.

5.1 IMPLEMENTATION PLATFORM / ENVIRONMENT

The HRMS chatbot was developed and tested on a local computing environment at Tech Elecon Pvt. Ltd., utilizing a CPU-only laptop to ensure a fully confidential setup as mandated by organizational policy. The system specifications are as follows:

- **Operating System**: Windows 11 (or specify your OS if different).
- **Processor**: Intel Core i5-10th Gen (or specify your CPU).
- **RAM**: 8GB (adjust based on your system).
- **Storage**: 256GB SSD (adjust as needed).
- **Python Version**: 3.9, chosen for compatibility with key libraries and tools.

The implementation relied on open-source software to maintain cost-effectiveness and local operation:

- **Ollama**: Facilitates local hosting and inference of lightweight NLP models, ensuring no external data transmission.
- **Streamlit**: Powers the web-based user interface, offering rapid development and interactive features.
- **ChromaDB**: Serves as a local vector database for storing and retrieving company handbook embeddings.
- **Development Tools**: Visual Studio Code for coding, Git for version control, and a Python environment with pip-managed dependencies.

This CPU-based setup was selected to align with the internship's resource constraints and confidentiality requirements, with plans to explore GPU-enabled systems for future performance enhancements. No cloud services were utilized, reinforcing data security.

5.2 PROCESS / PROGRAM / TECHNOLOGY / MODULES SPECIFICATION(S)

The implementation process involved a structured approach to building the HRMS chatbot, leveraging Retrieval-Augmented Generation (RAG) for accurate responses. Key steps and technologies include:

- **Data Preparation:**

The company handbook (PDF format) was processed into text and converted into embeddings using nomic-embed-text:latest.

These embeddings were stored in ChromaDB, creating a persistent local vector database (chroma directory) for similarity searches.

- **Core Program:**
  - **query_data.py:** Handles query processing:
    - **Input**: Accepts user queries via Streamlit or command-line arguments.
    - **Retrieval**: Performs similarity search on ChromaDB (k=5) to fetch relevant handbook snippets.
    - **Prompting**: Constructs a prompt:

```
You are an HR assistant chatbot for GBS Technologies.
Provide a concise, professional response relevant to HR policies, procedures, or information outlined in the handbook.
Answer the question based only on the given context below from the company handbook.
If there is even a little bit information related to question found than give that to the user.
Otherwise, don't try to make up an answer.
Act like you know these information and don't let user to know that you are using context from company handbook.
```

```
{context}

---

Question: {question}
```

- **Inference**: Uses Ollama to invoke models like llama3.2:1b for response generation.
- **Output**: Returns the response with source metadata for transparency.

  - **chatbot_ui.py:** Implements the Streamlit interface:
    - Features a fixed “Chatbot” header, chat history display, input field, and sidebar with timestamps, export options, and quick question buttons (e.g., “What time is it?”).

- **Technologies:**
  - **Ollama**: Hosts local models (qwen2.5:1.5b, deepseek-r1:1.5b, gemma3:1b, llama3.2:1b), each with sizes ranging from 815 MB to 1.3 GB.
  - **ChromaDB**: Manages handbook embeddings with nomic-embed-text:latest (274 MB).
  - **Streamlit**: Provides a dynamic, web-based UI with Python integration.
- **Modules:** Custom scripts include query_data.py for logic, populate_database.py for data ingestion, and chatbot_ui.py for the interface, all built on Python 3.9.

The process ensures a fully local workflow, from data storage to response generation, aligning with Agile methodology principles of iterative development and testing.

5.3 FINDING / RESULTS / OUTCOMES

The implementation yielded a functional HRMS chatbot with the following outcomes:

- **Functionality**: The chatbot successfully retrieves and responds to HR queries (e.g., “What’s the leave policy?”) based on the company handbook, displayed via Streamlit with interactive features.
- **Model Testing:**
  - **qwen2.5:1.5b (986 MB):** Fast inference, moderate quality, occasional verbosity.
  - **deepseek-r1:1.5b (1.1 GB):** Balanced speed and accuracy, suitable for concise answers.
  - **gemma3:1b (815 MB):** Quickest response time, simpler outputs with less depth.
  - **llama3.2:1b (1.3 GB):** High-quality, detailed responses, slightly slower on CPU.
- **Performance Metrics:**
  - Response times ranged from approximately 4-8 seconds per query on a CPU-only system (exact times depend on your observations).
  - Memory usage varied from 2-4 GB, fitting within the 8GB RAM constraint.
- **Interface Features**: The Streamlit UI includes chat history, timestamps, and export capabilities, enhancing usability for employees.
- **Challenges:**
  - CPU limitations slowed inference for larger models like llama3.2:1b.
  - Initial responses from deepseek-r1:1.5b included unwanted tags (e.g., <think>), resolved through prompt refinement.

These findings demonstrate the chatbot’s ability to deliver secure, handbook-based responses locally, laying a foundation for future enhancements.

5.4 RESULT ANALYSIS / COMPARISON / DELIBERATIONS

- **Model Comparison:**

Table 5.1 Model Performance Comparison

Model	Size	Response Time (min)	Memory (GB)	Quality (1-5)	Remarks
qwen2.5:1.5b	986 MB	~1-3	~3-3.5	3.5	Fast, slightly verbose
Deepseek-r1:1.5b	1.1 GB	~1-3	~3-4	4.0	Balanced, tag issues fixed
gemma3:1b	815 MB	~1-2	~3.0	3.0	Quick, less detailed
llama3.2:1b	1.3 GB	~1-3	~3.5-4	4.5	High quality, slower

- **Analysis:**
  - **Speed**: gemma3:1b excelled due to its smaller size, while llama3.2:1b lagged slightly, reflecting a trade-off between model complexity and inference speed on CPU.
  - **Quality**: llama3.2:1b provided the most detailed and accurate responses, ideal for professional HR use, whereas gemma3:1b offered simpler outputs.
  - **Resource Use**: All models operated within the 8GB RAM limit, with gemma3:1b being the most efficient and llama3.2:1b pushing the upper boundary.
- **Deliberations:**
  - **Current Setup**: deepseek-r1:1.5b and llama3.2:1b strike a balance for CPU use, with prompt tuning mitigating initial issues. gemma3:1b suits low-resource scenarios.
  - **Future Optimization**: Transitioning to a GPU-enabled system could reduce response times to 1-3 seconds, supporting larger models for enhanced quality.

- **Next Steps**: Planned features like user authentication, persistent chat history, and feedback analysis will build on this foundation, requiring further testing and integration .

The results validate the local implementation’s feasibility while highlighting areas for scalability and refinement

6. TESTING

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